Project: Worked examples
Title: Sealed heating system: emitters and expansion vessel
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A closed LTHW system below 100 °C: the catalogue output (at ΔT 50 K) each room's emitter needs at the design flow, return and room temperatures, checked against the selected emitters; and the minimum expansion vessel for the system volume, fill and maximum temperatures, static head and pressures. Emitter temperature difference: ΔT = (t_f + t_r)/2 − t_i, or the log mean (t_f − t_r)/ln((t_f − t_i)/(t_r − t_i)) Output at design: Φ = Φ50·(ΔT/50)^n; required catalogue output Φ50 = Φ_req/(ΔT/50)^n System volume: V_s = Σ V_em + V_pipe + V_plant Expansion: e_x = ρ(t_fill)/ρ(t_max) − 1 (water density by Kell 1975), or a glycol record Pre-charge: p_i = ρ·g·h_static + margin (≥ static head) Vessel: V_e = e_x·V_s/[1 − (p_i + 1)/(p_f + 1)] (pressures in bar gauge, + 1 bar for absolute)

Ref: BS EN 442-2:2014 cl. 5 — Radiators and convectors — test methods and rating (characteristic equation)
Ref: CIBSE Guide B1 2016, §1.4 — Heating — sealed systems and expansion
Ref: BS EN 12828:2012+A1:2014 cl. 4.6 — Heating systems in buildings — expansion vessels and pressure

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
rooms	Rooms and emitters	5 row(s)		evidence: Room heat loss calculation (B4) and radiator survey	any
rooms[1]		name = Lounge; Phi = 1800 W; Phi50 = 3600 W; V_em = 12 L			
rooms[2]		name = Kitchen; Phi = 1100 W; Phi50 = 2200 W; V_em = 8 L			
rooms[3]		name = Bed 1; Phi = 900 W; Phi50 = 1800 W; V_em = 7 L			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
rooms[4]		name = Bed 2; Phi = 700 W; Phi50 = 1400 W; V_em = 5 L			
rooms[5]		name = Bathroom; Phi = 600 W; Phi50 = 1100 W; V_em = 4 L			
t_f	Flow temperature	55	degC	spec: Heat pump design flow temperature	30 … 90 degC
t_r	Return temperature	45	degC	spec: Heat pump design return temperature	30 … 90 degC
t_i	Room temperature	21	degC	spec: Living-room design temperature	16 … 24 degC
n	Emitter exponent	1.3		assumption: Panel radiators, manufacturer exponent about 1.3.	1.1 … 1.5
dT_method	Mean temperature difference	arithmetic mean (catalogue rating)		assumption: Catalogue ratings use the arithmetic mean.	any
V_pipe	Pipework water content	40	L	assumption: 15 mm and 22 mm copper, about 120 m.	0 … 10000 L
V_plant	Plant water content (boiler, cylinder coil, buffer)	15	L	spec: Heat pump and plate exchanger content, datasheet	0 … 10000 L
t_fill	Fill temperature	10	degC	assumption: Mains water at fill.	4 … 30 degC
t_max	Maximum system temperature	60	degC	spec: Heat pump maximum flow temperature	30 … 99 degC
h_static	Static height above the vessel	7	m	drawing: M-SEC-001 rev A	0 … 50 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_margin	Pre-charge margin over static head	0.3	bar	assumption: Usual margin over static head.	0 … 1 bar
p_f	Final (maximum working) pressure, gauge	2.5	bar	assumption: 0.5 bar below the safety valve.	0.5 … 6 bar
p_prv	Safety valve setting, gauge	3	bar	spec: Heat pump safety valve	1.5 … 6 bar
V_vessel	Selected vessel volume	12	L	evidence: Site survey: existing 12 L vessel	1 … 5000 L

3. Assumptions

A1. Closed LTHW system below 100 °C; emitter output follows Φ50·(ΔT/50)^n with the manufacturer's n. (Catalogue ratings at 75/65/20 °C.)
A2. Water density from the Kell (1975) correlation, within about 0.01 % of IAPWS-95 at atmospheric pressure, 0–100 °C; glycol mixtures take e from a record. (Formula, not a table.)
A3. Gauge pressures are made absolute by adding 1 bar, as the vessel relation is usually written. (Within 1.3 % of the standard atmosphere.)
A4. n (Emitter exponent) is taken as 1.3. (Panel radiators, manufacturer exponent about 1.3.)
A5. dT_method (Mean temperature difference) is taken as arithmetic mean (catalogue rating). (Catalogue ratings use the arithmetic mean.)
A6. V_pipe (Pipework water content) is taken as 40 L. (15 mm and 22 mm copper, about 120 m.)
A7. t_fill (Fill temperature) is taken as 10 degC. (Mains water at fill.)
A8. p_margin (Pre-charge margin over static head) is taken as 0.3 bar. (Usual margin over static head.)
A9. p_f (Final (maximum working) pressure, gauge) is taken as 2.5 bar. (0.5 bar below the safety valve.)

4. Working

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[14] dT = (t_f + t_r) / 2 - t_i                                  BS EN 442-2 cl. 5
= (55 degC + 45 degC) / 2 - 21 degC
= 29 K
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Mean water-to-room temperature difference

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[15] dT_fr = t_f - t_r
= 55 degC - 45 degC
= 10 K
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Flow-to-return temperature difference (what the water gives up across the emitters)

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[16] t_m = (t_f + t_r) / 2
= (55 degC + 45 degC) / 2
= 50 degC
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Mean water temperature, (flow + return)/2

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[17] f_dT = (dT / 50 K)^n
= (29 K / 50 K)^1.3
= 0.49256
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Output correction from ΔT 50 K

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[18] R1 = □_req/f_dT
= 1800 / 0.49256
= 3654.4 W
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Lounge: catalogue output needed; selected 3600 W gives 1773.2 W (utilisation 1.02)

[19] R2 = Φ_req/f_dT
= 1100 / 0.49256
= 2233.2 W
Kitchen: catalogue output needed; selected 2200 W gives 1083.6 W (utilisation 1.02)
[20] R3 = Φ_req/f_dT
= 900 / 0.49256
= 1827.2 W
Bed 1: catalogue output needed; selected 1800 W gives 886.6 W (utilisation 1.02)
[21] R4 = Φ_req/f_dT
= 700 / 0.49256
= 1421.2 W
Bed 2: catalogue output needed; selected 1400 W gives 689.58 W (utilisation 1.02)
[22] R5 = Φ_req/f_dT
= 600 / 0.49256
= 1218.1 W
Bathroom: catalogue output needed; selected 1100 W gives 541.81 W (utilisation 1.11)
[23] Phi_tot = Σ Φ_req
= 5100 W
= 5100 W
Total heat required by the rooms
[24] V_em = Σ V_em
= 36 L
= 36 L
Emitter water content
[25] V_s = V_em + V_pipe + V_plant
= 36 L + 40 L + 15 L
= 91 L
System water volume
[26] rho_fill = ρ(t_fill), Kell 1975
= t = 10 °C
= 999.7 kg/m³
Water density at fill
[27] rho_max = ρ(t_max), Kell 1975
= t = 60 °C
= 983.2 kg/m³
Water density at maximum temperature
[28] e_x = rho_fill / rho_max - 1
= 999.7 kg/m³ / 983.2 kg/m³ - 1
= 0.016783
Expansion coefficient of water
[29] p_st = rho_fill * 9.80665 m/s^2 * h_static
= 999.7 kg/m³ × 9.80665 m/s^2 × 7 m
= 0.68626 bar
Static head at the vessel
[30] p_i = p_st + p_margin
= 0.68626 bar + 0.3 bar
= 0.98626 bar
Pre-charge (initial) pressure, gauge
[31] V_e = e_x * V_s / (1 - (p_i + 1 bar) / (p_f + 1 bar)) CIBSE Guide B1 §1.4
= 0.016783 × 91 L / (1 - (0.98626 bar + 1 bar) / (2.5 bar + 1 bar))
= 3.5312 L
Minimum expansion vessel volume

[32] u_emit = max over rooms □_req/(□50·f_dT)
= worst room
= 1.1074
Emitter utilisation (governing room)
[34] u_vessel = V_e / V_vessel
= 3.5312 L / 12 L
= 0.29427
Vessel utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
dT	29 K	Mean temperature difference
dT_fr	10 K	Flow-to-return temperature difference
t_m	50 degC	Mean water temperature
f_dT	0.49256	Output correction factor
R1	3654.4 W	Lounge: Φ50 needed
R2	2233.2 W	Kitchen: Φ50 needed
R3	1827.2 W	Bed 1: Φ50 needed
R4	1421.2 W	Bed 2: Φ50 needed
R5	1218.1 W	Bathroom: Φ50 needed
Phi_tot	5100 W	Total heat required
V_em	36 L	Emitter water content
V_s	91 L	System water volume
e_x	0.016783	Expansion coefficient
p_i	0.98626 bar	Pre-charge pressure
V_e	3.5312 L	Minimum vessel volume
u_emit	1.1074	Emitter utilisation
u_vessel	0.29427	Vessel utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Selected emitters against room heat (governing room)	u_emit = 1.1074	u_emit ≤ 1	1.11	
PASS	Selected vessel against minimum volume	u_vessel = 0.29427	u_vessel ≤ 1	0.294	BS EN 12828 cl. 4.6

7. Applicability

WARNING — record.n: n: a tabulated value not taken from a reference record (Emitter exponent); enter it as one so its source, edition and licence travel with it.

8. Verification

Template bs.sealed-system@1.1.0 Content hash d3a2350bd0878328 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0